

Jay Rao

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EDUCATION

Northwestern University

Bachelor of Arts (B.A.) in **Computer Science**

Expected December 2027

Concentration in **Artificial Intelligence**

SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Node.js, React, PyTorch, NumPy, Pandas, Playwright

Tools & Platforms: AWS, Terraform, Git, GitHub, Linux/Bash, Jupyter Notebook, Render, Vercel

EXPERIENCE

Founding Engineer | GridGate LLC

Feb. 2026 - Present

- Sole engineer on GridGate Queue Analysis v1.0, a full MISO queue intelligence system: 19-source data lake (20+ scrapers), 9-module feature pipeline, and two CatBoost models trained on 3,000+ historical projects
- Built a two-stage ML pipeline: success/withdrawal classifier (ROC-AUC 0.93, 85% accuracy) and 5-class withdrawal phase classifier (74% accuracy, one-vs-rest ROC-AUC 0.94), both with SHAP explainability
- Automated collection of 100,000+ county meeting minutes PDFs across 700+ MISO counties (15 states, 2018–present) via Legistar, CivicPlus, custom crawlers, and Claude agent discovery for non-standard sites
- Architected a location-aware social sentiment pipeline ingesting Facebook, X, Reddit, YouTube, Google Maps, and Change.org with geo-tagged city/county/state resolution and weighted relevance scoring

Product Manager | iloveyou LLC

Feb. 2026 - Present

- Lead developer of an early-stage consumer social media app, managing 3 junior developers across sprint planning, task assignment, and code reviews, towards a hard August 5th launch at 4 flagship universities
- Built the full backend from scratch with Node.js/TypeScript, Fastify, PostgreSQL, JWT auth with Apple Sign-In and email fallback, and REST APIs covering feed, friends, messaging, events, and groups
- Implemented the full iOS frontend in SwiftUI (MVVM), including onboarding, fruit community assignment, spinning wheel fruit reveal, profile creation, and all core functionality (feed, messaging, events, and groups)
- Engineered all AWS infrastructure via Terraform (VPC, ECS Fargate, ALB, RDS, S3, IAM) and developed the full CI/CD pipeline via GitHub Actions

Computer Science Educator | Northwestern TIDAL & tiilt labs

Jan. 2026 - Jun. 2026

- Co-led two fifth-grade classrooms in the Evanston/Skokie School District (D65) each week, guiding 40+ students through creative, project-based Python curriculums using TunePad and SportSense

PROJECTS

vibes.fm | *JavaScript, React, Node.js, AWS, MySQL, Transformers.js, OAuth 2.0*

- Engineered a full-stack AI playlist generator that encodes track metadata and lyrics into vector embeddings using a local Hugging Face sentence-transformer model
- Implemented an in-memory vector cache and cosine similarity search to match natural language user “vibe” prompts to songs, automatically exporting matched playlists directly to Spotify via PKCE OAuth 2.0

ImageLab | *Python, TypeScript, SQL, FastAPI, React, Next.js, AWS, Tailwind CSS, Pillow, Puter.js*

- Developed a highly scalable image storage and processing pipeline featuring 15+ programmatic transformations, secure JWT authentication, and automated AI image labeling via Amazon Rekognition
- Architected the cloud infrastructure with S3 and RDS to support efficient media storage and data retrieval

PitchSense | *Python, YOLOv5, OpenCV, PyTorch, scikit-learn, NumPy, Pandas*

- Built a computer vision pipeline using YOLOv5 and ByteTrack for object detection with optimized memory management, K-Means clustering for team identification via jersey color analysis, View Transformer for 2D tactical views with Optical Flow camera compensation and continuous ball tracking using Pandas

nuPython | *C, C++, STL*

- Engineered a complete interpreter, memory management (RAM), and GDB-style debugger for nuPython across 2,000+ lines of C/C++, supporting integers, floats, strings, booleans, pointers, and None types